



Homework/Programming Assignment #1

Homework Due: 02/12/2024- 5:00PM

Name/EID:

Email:

Signature (*required*)

I/We have followed the rules in completing this Assignment.

Name/EID:

Email:

Signature (*required*)

I/We have followed the rules in completing this Assignment.

Question	Points	Total
HA 1		
HA 2		
HA 3a		
HA 3b		
HA 4a		
HA 4b		
HA 4c		
HA 5a		
HA 5b		
HA 6		
HA 7		
HA 8		
HA 9		
PA 1		
PA 2		
PA 3		

Instruction:

1. Remember that this is a graded homework assignment. It is the equivalent of a **mini take-home exam**.
2. You are to work **alone** or **in teams of two** and are **not to discuss the problems** with anyone other than the TAs or the instructor.
3. It is open book, notes, and web. But you should cite any references you consult.
4. **You may use large language models (LLMs) such as ChatGPT or Gemini. If you do so, you must clearly state: “I used ChatGPT to assist in responding to this assignment.” In addition, you must include the prompts you used and briefly describe how you validated the model’s responses. The full sequence of prompts should be appended to your submission.**
5. Unless I say otherwise in class, **it is due before the start of class on the due date** mentioned in the P/H Assignment.
6. **Sign and append** this score sheet as the first sheet of your assignment.
7. Remember to submit your assignment in Canvas.



➤ Homework Assignment (HA)

1. Prove that a rotation matrix $\mathbf{R} \in \mathbf{SO}(3)$ is a rigid body transformation.
2. Prove that rigid transformation $\mathbf{g} = (\mathbf{p}, \mathbf{R})$ on a vector $\mathbf{v} = \mathbf{s} - \mathbf{r}$ is $\mathbf{g}_*(\mathbf{v}) := \mathbf{R}\mathbf{v}$.
3. (MLS Book [1]) *Properties of rotation matrices*

Let $R \in \mathbf{SO}(3)$ be a rotation matrix generated by rotating about a unit vector ω by θ radians. That is, R satisfies $R = \exp(\hat{\omega}\theta)$.

- a. Show that the eigenvalues of $\hat{\omega}$ are 0, i , and $-i$, where $i = \sqrt{-1}$. What are the corresponding eigenvectors?
 - b. Show that the eigenvalues of R are 1, $e^{i\theta}$, and $e^{-i\theta}$. What is the eigenvector whose eigenvalue is 1? What is the physical interpretation of this eigenvector?
4. (MLS Book [1]) *Properties of skew-symmetric matrices*

Show That the following properties of skew- symmetric matrices are true:

- a. If $R \in \mathbf{SO}(3)$ and $\omega \in \mathbb{R}^3$, then $R\hat{\omega}R^T = \widehat{R\omega}$.
- b. If $R \in \mathbf{SO}(3)$ and $v, \omega \in \mathbb{R}^3$, then $R(v \times \omega) = (Rv) \times (R\omega)$.

5. (MLS Book [1]) *Unit quaternions*

Let $Q = (q_o, \vec{q})$ and $P = (p_o, \vec{p})$ be quaternions, where $q_o, p_o \in \mathbb{R}$ are the scalar parts of Q and P and $\vec{q}, \vec{p}$ are the vector parts.

- a. Show that the set of *unit* quaternions satisfies the axioms of the group.
- b. Let x be a point and let X be a quaternion whose scalar part is zero and whose vector part is equal to x (such a quaternion is called a *pure* quaternion). Show that if Q is a unit quaternion, the product QXQ^* is a pure quaternion and the vector part of QXQ^* satisfies

$$(q_o^2 - \vec{q} \cdot \vec{q})\vec{x} + 2(q_o (\vec{q} \times \vec{x}) + (x \cdot \vec{q})\vec{q})$$

6. Recall from notes or the book that the explicit matrix representations for the rigid body rotations:
 - $R_x(\phi) : \mathbb{R} \mapsto \mathbf{SO}(3)$ corresponding to a rotation of ϕ radians about the x-axis — i.e. a 3×3 matrix whose elements contain expressions such as $\sin(\phi)$ and $\cos(\phi)$
 - $R_y(\theta) : \mathbb{R} \mapsto \mathbf{SO}(3)$ corresponding to a rotation of θ radians about the y-axis.
 - $R_z(\psi) : \mathbb{R} \mapsto \mathbf{SO}(3)$ corresponding to a rotation of ψ radians about the z-axis.

In the class, we have learned that the ZXZ and ZYZ are the most common (the explicit form of the ZYZ Euler angles is in MLS). Here, construct the explicit representation for $R_{xyz}(\psi) : \mathbb{R} \mapsto \mathbf{SO}(3)$ given by



$$R_{xyz}(\psi, \theta, \phi) = R_x(\phi)R_y(\theta)R_z(\psi)$$

Construct the inverse function $R_{xyz}^{-1} : SO(3) \mapsto \mathbb{R}^3$ such that $\forall R \in SO(3)$ if $y = R_{xyz}^{-1}(R)$ then $R = R_{xyz}(y)$, i.e., derive the formulas such as in (2.20) in MLS.

7. **Exercise 3.16** (pg.121) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [2].
8. **Exercise 3.18** (pg.123) in *Modern Robotics: Mechanics, Planning, and Control* (Lynch et al.) [2].
9. Consider the pelvic osteotomy situation illustrated in **Fig. 1**. Here we assume that a three locating pins have been inserted into the patient's pelvis, and that a CT scan of the pelvis with the pins inserted has been produced. The patient has been placed onto the operating table. Also, a magnetic navigation system (here, the Northern Digital Aurora) is present in the room.

Two surgical tools are available:

- A probe/pointer device
- An osteotome (essentially a fancy chisel) that will be used to cut the pelvis.

6-DOF Aurora tracking sensors have been attached to the handle of each tool and an additional 6-DOF sensor has been affixed rigidly to the pelvis. The Aurora is capable of determining the position and orientation of each sensor relative to the Aurora base unit.

We will define the following coordinate systems:

$\mathbf{F}_B$ = Coordinate system of tracking system base unit

$\mathbf{F}_D$ = Coordinate system of tracking device on pointer handle

$\mathbf{F}_H$ = Coordinate system of tracking device on osteotome handle

$\mathbf{F}_G$ = Coordinate system of tracking device attached to pelvis

$\mathbf{F}_C$ = Coordinate system of CT image

We also have the following relationships

$\mathbf{F}_{Bx}$ = Measured 6 DOF pose of tracking device x relative to base unit

$\mathbf{F}_{HK}$ = 6 DOF pose of osteotome blade relative to osteotome handle tracking device

$\mathbf{F}_{DK}$ = 6 DOF pose of pointer tip relative to pointer handle tracking device

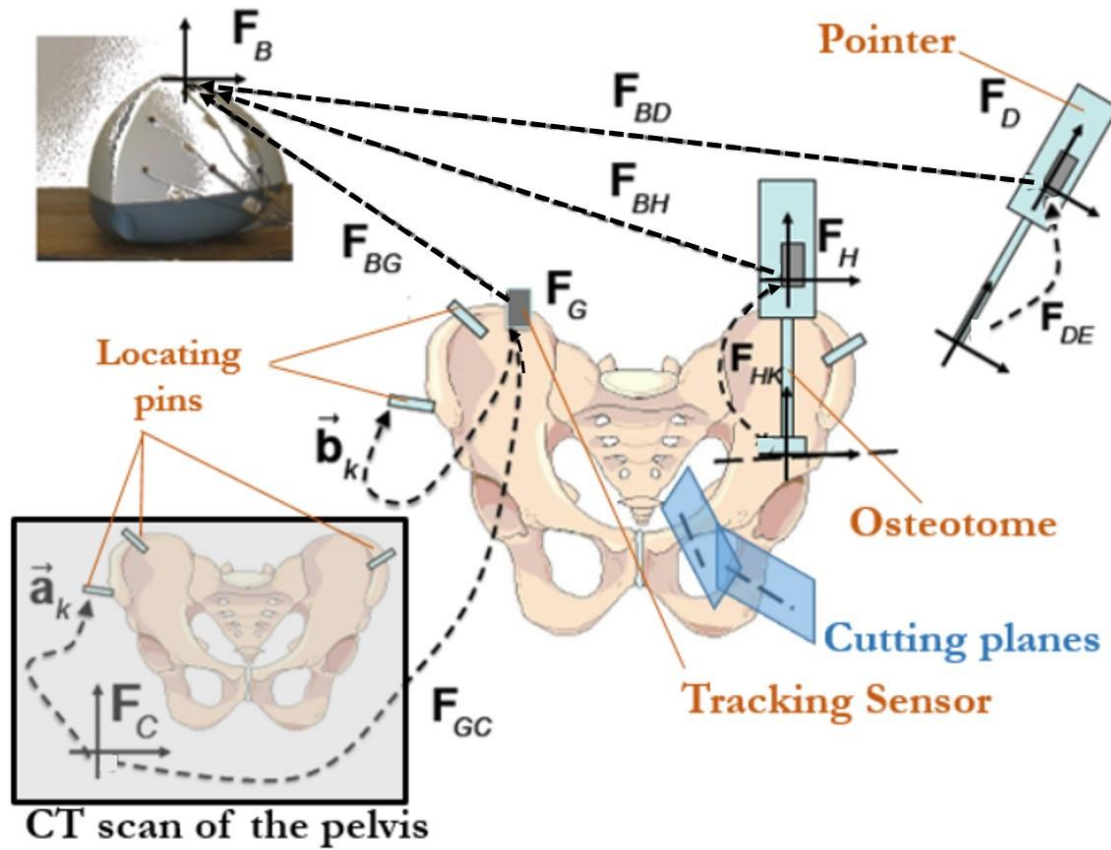


Fig. 1: Computer-Assisted Osteotomy

$\mathbf{a}_k$ = Position of the top of pin k in CT coordinates

$\mathbf{b}_k$ = Position of the top of pin k relative to tracking device G

Suppose that we have touched the tops of the three fiducial pins and used the results to compute a registration transformation $\mathbf{F}_{GC}$ such that $\mathbf{F}_{GC} \mathbf{a}_k = \mathbf{b}_k$. Give an expression for computing the position and orientation $\mathbf{F}_{CK}$ of the osteotome blade in CT coordinates, based on the available tracking system measurements $\mathbf{F}_{Bx}$ [3].



➤ Programming Assignment (PA)

1. Write functions that given a rotation matrix $R \in SO(3)$ returns:
 - a. Its equivalent axis-angle representation.
 - b. Quaternion representation.
 - c. ZYZ and roll-pitch-yaw representation.
2. Write functions that:
 - a. Given an axis-angle representation returns the equivalent rotation matrix.
 - b. Given a quaternion representation returns the equivalent rotation matrix.
3. Using the functions you have written write a program that allows the user to specify an initial configuration of a rigid body by T , a screw axis specified by $\{q, \hat{s}, h\}$ in the fixed frame $\{s\}$, and the total distance traveled along the screw axis θ . The program should calculate the final configuration $T_1 = e^{[\mathcal{S}]\theta}T$ attained when the rigid body follows the screw $\mathcal{S}$ a distance θ , as well as the intermediate configurations at $\frac{\theta}{4}, \frac{\theta}{2}$, and $\frac{3\theta}{4}$. At the initial, intermediate, and final configurations, the program should plot the $\{b\}$ axes of the rigid body. The program should also calculate the screw axis $\mathcal{S}_1$ and the distance θ_1 following $\mathcal{S}_1$ that takes the rigid body from T_1 to the origin and it should plot the screw axis $\mathcal{S}_1$. Test the program with $q = (0, 2, 0)$, $\hat{s} = (0, 0, 1)$, $h = 2$, $\theta = \pi$, and

$$T = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

➤ References:

1. (MLS Book) Murray, R.M., Li, Z., Sastry, S.S., “*A Mathematical Introduction to Robotic Manipulation.*”, **Chapter 2**.
2. Lynch and Park, “*Modern Robotics*,” Cambridge U. Press, 2017, **Chapter 3**.
3. Computer Integrated Surgery course, Russell H. Taylor, JHU